

Quantum Mechanics

1. Determine the wavelength of an electron moving with the speed of 5.41×10^5 m/s.

(Ans 1.37×10^{-9} m)

2. Determine the wavelength of the particle. Consider an electron with velocity at 10^7 cm/sec

(Ans 7.27×10^{-11} m)

3. What is de broglie wavelength if a photon with momentum of 4.4×10^{-34} kg.m/s?

(Ans 1.5 m)

4. The momentum of photon A is p_a and momentum of photon B is p_b . If $p_b = 1/3 p_a$ then what is the relation between the wavelengths?

(Ans $\lambda_a = (1/3) \lambda_b$)

5. Calculate the de Broglie wavelength of a golf ball whose mass is 40 grams and whose velocity is 6 m/s.

(Ans $= 2.76 \times 10^{-33}$ m)

6. What is the frequency of the photon whose energy is 66.3 eV?

(Ans $= 1.6 \times 10^{16}$ Hz)

7. A photon of energy 6×10^{-20} J has linear momentum of 2×10^{-30} kg.m/s. Verify this statement.

1) If the uncertainty in position of an electron is 4×10^{-10} m. Calculate uncertainty in its momentum.

(Ans: 2.62×10^{-25} kg m/sec)

2) A bullet of mass 0.050 kg is moving with a velocity of 800 m/s. The speed of bullet measured accuracy 0.01%. Calculate the accuracy with which the position of the bullet can be located.

(Ans: 2.62×10^{-32} m)

3) Calculate the minimum uncertainty in measurement of momentum of an electron if the uncertainty in locating it is $1 \text{ \AA}$.

(Ans: 1.055×10^{-22} kg m/sec)

4) The uncertainty in the location of a particle is equal to its de Broglie wavelength. Calculate the uncertainty in its velocity.

5) An electron and a 150 gm baseball are travelling at 220 m/sec measured to an accuracy of 0.065 %. Calculate the uncertainties in the position of each of the bodies. Compare and comment on the results.

(Ans: uncertainty of electron is greater than baseball)

6) An electron has a momentum 5.4×10^{-26} kg-m/sec with an accuracy of 0.05%. Find the minimum uncertainty in the location of electron.

(Ans: $3.9 \text{ \mu m}$)

8. Find phase and group velocities of electron whose de Broglie wavelength is $1.2 \text{ \AA}$.

(Ans: $v_p = 3.036 \times 10^6$ m/s, $v_g = 6.06 \times 10^6$ m/s)

9. The electrons which are at rest accelerated through a potential difference of 250 V. calculate

- I. The velocity of electron
- II. Phase velocity of electron
- III. De Broglie wavelength, momentum, wave number

(Ans: $v = 9.376 \times 10^6$ m/s, $v_p = 9.59 \times 10^9$ m/s, $\lambda = 7.78 \times 10^{-11}$ m, $p = 8.532 \times 10^{-24}$ kg m/s, $n = 1.28 \times 10^{10}$ /m)

10. Calculate the energy of the lowest three levels for an electron in a square well of width $2\text{\AA}$.

(Ans: $E_1 = 15.1 \times 10^{-19} \text{J}$

$E_2 = 60.4 \times 10^{-19} \text{J}$

$E_3 = 135.9 \times 10^{-19} \text{J}$)

11. Find the energy of an electron moving in one dimension in an infinitely high potential well of width $1\text{\AA}$. Also calculate the energy difference between the ground state and the first excited state.

(Ans: energy = $n^2 \times 6.038 \times 10^{-18} \text{J}$

Energy difference = $18.114 \times 10^{-18} \text{J}$)

12. Find the lowest energy of a neutron within a nucleus of dimension 10^{-14}m . (the mass of neutron = $1.67 \times 10^{-27} \text{kg}$)

(Ans: $3.29 \times 10^{-13} \text{J}$)

13. The lowest energy of an electron trapped in a potential well is 38eV . Calculate the width of the well.

(Ans: Width = $0.9965 \text{\AA}$)

14. A particle is confined to a one-dimensional box of width $25 \text{\AA}$ and infinite height. Calculate the probability of finding the particle within an interval of $5\text{\AA}$ at the center of the box in its least energy state.

(Ans: probability = 20%)